

Docker Container Not Starting SOP

Objective

Restore service availability when a Docker container exits immediately after the run command or fails during startup.

Scope

Applicable to Docker-based services, containerized application deployments, DevOps teams, SRE teams, platform teams, and synthetic incident workflows across development, staging, and production-like environments.

Pre-requisites

- 1. Access to Docker host, Docker CLI, container runtime logs, and the target container or image name.
- 2. Permission to inspect containers, images, volumes, networks, resource limits, and compose files where applicable.
- 3. Access to application logs, deployment artifacts, Dockerfile, environment variables, and registry credentials.
- 4. Monitoring access for host CPU, memory, disk, container restart count, and container health checks.
- 5. Escalation contact for application, DevOps, security, registry, infrastructure, and platform teams.

Incident Metadata

Field	Value
Ticket ID	T001
Issue	Docker container not starting
Category	Deployment
Service	Docker
Severity	High
Status	Resolved
Description	Container exits immediately after run command
Expected Resolution	Check logs using docker logs and fix missing environment variables

Related Log Evidence

Log ID	Timestamp	Level	Message	Error Code
N/A	N/A	N/A	No direct log evidence found in logs CSV for this ticket.	N/A

Procedure

- 1. Confirm the impacted container name, image tag, command used, environment, and first failure timestamp.
- 2. Run `docker ps -a` to confirm the exit status and restart count.
- 3. Review container logs to identify missing environment variables, dependency failures, command errors, or application crashes.
- 4. Inspect the container configuration, entrypoint, command, mounted files, ports, and environment variables.
- 5. Validate required secrets, config files, volumes, and network dependencies are available on the host.
- 6. Run the image locally with a shell or debug command to isolate startup failure.
- 7. Fix the missing variable, command, dependency, permission, or configuration issue.
- 8. Recreate the container and confirm status is Up and application health checks pass.

Common Commands

```
docker ps -a
docker logs <container-name-or-id> --tail=100
docker inspect <container-name-or-id>
docker stats --no-stream
docker images
docker network ls && docker network inspect <network-name>
docker volume ls && docker volume inspect <volume-name>
docker system df
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Container exits immediately	Missing environment variable	Add required variable or secret
Exit code 1	Application startup failure	Check logs and fix app configuration
Entrypoint not found	Invalid Dockerfile command	Correct ENTRYPOINT or CMD
Health check failing	App not ready or wrong port	Fix health check path, port, or startup timing

Best Practices

1. Define mandatory environment variables in deployment templates.
2. Use health checks for long-running services.
3. Avoid hardcoded local paths in container startup commands.
4. Keep a debug run command for production-like images.

References

Docker Docs | Internal Tickets CSV | Internal Logs CSV | Container Runtime Logs | Monitoring Dashboards | DevOps Runbook